

Final Analysis Insight of Assignment 1

Project Title:

PRODUCT INNOVATION HUB

Introduction: In this report Analyses Product Data set.

Key fields: *Product ID, Product Name, Brand Name, Quantity, Category, Price Range, Day, Country Code, Month*

- **Data Summarization:**

SUM, COUNT AND AVERAGE FUNCTIONS

1) SUM FUNCTION: In this function we using calculation of totals, subtotals and summarizing data.

Syntax: **FORMULAE: =SUM (D2:D35)**

Screenshot of Sum Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=SUM(Dataset[Price ($)])` for cell M9. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 8 rows of data.

	A	B	C	D	E	F	G	H	I	J
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
1	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
2	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
3	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
4	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
5	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
6	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
7	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK

Answer: **Total Price of All Products: \$10,100**

2) COUNT FUNCTION: Count function is always counting only Numeric value and Integer and Decimal value. The count Function not string text format.

Syntax: FORMULAE: =COUNTA (B2:B35)

Screenshot of Count A Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=COUNTA(Dataset[Product Name])` in cell M17. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 10 rows of data.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Answer: Total Number of Products: 34

- 3) **AVERAGE FUNCTION:** Count function is always counting only Numeric value and Integer and Decimal value. The count Function not string text format.

Syntax: FORMULAE: =AVERAGE (D2:D35)

Screenshot of Average Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=AVERAGE(Dataset[Price ($)])` in cell M32. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 10 rows of data.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Answer: Average Price Of The Products: \$297

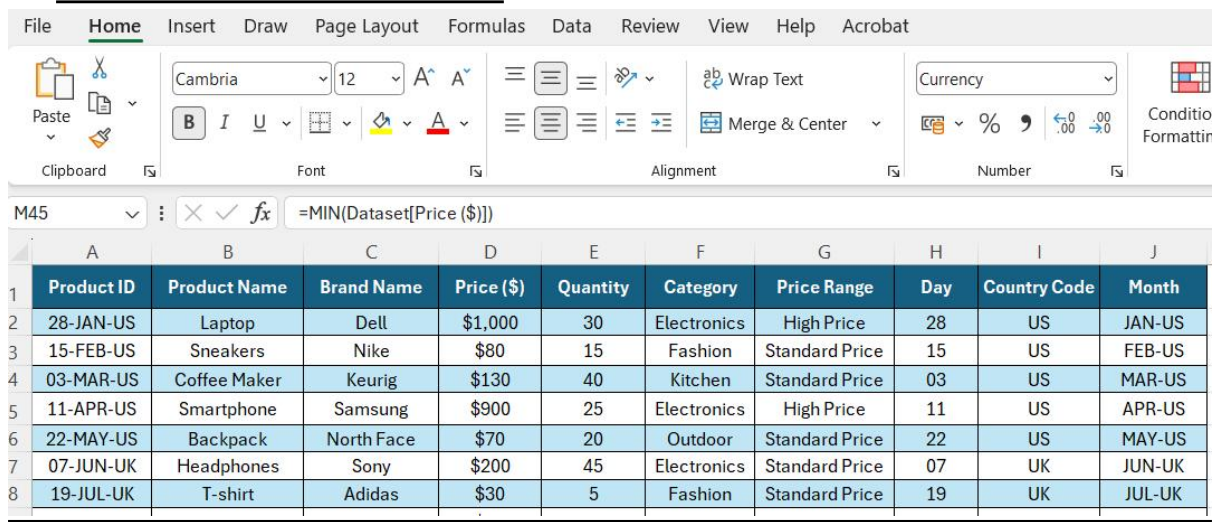
- **Data Range Analysis:**

MIN AND MAX FUNCTIONS

4) MIN FUNCTION: Calculating minimum value of Price Range.

Syntax: FORMULAE: =MIN (D2:D35)

Screenshot of MIN Table:



The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=MIN(Dataset[Price ($)])` in cell M45. The ribbon is set to the 'Home' tab. Below the ribbon, a table with 10 columns and 8 rows is visible. The columns are labeled: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The data rows contain various product information, including items like a Laptop, Sneakers, Coffee Maker, Smartphone, Backpack, Headphones, and T-shirt, with their respective prices and quantities.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK

Answer: Minimum Price of All Products: \$30

5) MAX FUNCTION: Calculating maximum value of Price Range.

Syntax: FORMULAE: =MAX (D2:D35)

Screenshot of MAX Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=MAX(Dataset[Price ($)])`. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 8 rows of data.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK

Answer: Maximum Price of All Products: \$1,000

• Logical Function:

IF FUNCTION

6) **IF FUNCTION:** It is fundamental tool. Performing comparisons between test result TRUE or FALSE.

Syntax: **FORMULAE:** `=IF(D2>=500,"High Price", IF(D2<500,"Standard Price", "True"))`

Screenshot of NESTED IF Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula `=IF(D2>=500,"High Price",IF(D2<500,"Standard Price","True"))`. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 8 rows of data.

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month	
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US	
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US	
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US	
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US	
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US	
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK	
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK	

G
Price Range
High Price
Standard Price
Standard Price
High Price
Standard Price
Standard Price
Standard Price
Standard Price
High Price

Answer:

Syntax: FORMULAE: =IF(D11>500,"High Price", "Standard Price")

Screenshot of IF Table:

The screenshot shows the Microsoft Excel interface. The formula bar at the top displays the formula: `=IF(D11>500,"high Price","Standard Price")`. Below the formula bar is a table with 10 columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The table contains 12 rows of data. The 'Price Range' column (G) shows 'High Price' for rows 1, 5, 10, and 12, and 'Standard Price' for rows 2, 3, 4, 6, 7, 8, 9, and 11.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK
11	14-OCT-UK	Hiking Boots	Timberland	\$130	10	Outdoor	Standard Price	14	UK	OCT-UK
12	17-JUN-IN	Laptop	HP	\$950	25	Electronics	High Price	17	IN	JUN-IN

Answer: Price Range: "Standard Price"

- **Condition Functions:**

SUMIF AND COUNTIF FUNCTIONS

7) **SUMIF FUNCTION:** SUMIF allows you to sum value. You need apply range criteria, sum range. Specified range are Summed.

Syntax: FORMULAE: =SUMIF (F2:F35,"Electronics", D2:D35)

Screenshot of SUMIF Table:

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK

Answer: Total Electronics category: \$8,050

8) **COUNTIF FUNCTION:** Counts the number of “Occurrence” that’s satisfy a given condition. Key point is Conditional counting.

Syntax: FORMULAE: =COUNTIF (D2:D35,"<100")

Screenshot of COUNTIF Table:

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month	
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US	
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US	
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US	
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US	
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US	
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK	
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK	
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK	

Answer: Number of Products with a Price less than: \$11

• Text Functions:

LEFT, RIGHT AND MID FUNCTIONS

9) **LEFT FUNCTION:** Can be used limit the number of characters displayed from left side of text string.

Syntax: **FORMULAE:** =LEFT(A2,2)

Screenshot of LEFT Table

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

H
Day
28
15
03
11
22
07
19
23
05

Answer: **Day: 28**

10) **RIGHT FUNCTION:** Can be used the number of characters displayed from right side of text string.

Syntax: **FORMULAE:** =RIGHT(A2,2)

Screenshot of RIGHT Table

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Country Code
US
US
US
US
US
UK
UK
UK
UK

Answer: Country Code: US

11) **MID FUNCTION:** Extract a sub string from middle of text string.

Syntax: FORMULAE: =MID(A2,4,6)

Screenshot of MID Table

The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The ribbon includes options for Clipboard, Font, Alignment, Number, and Condition Formatting. The formula bar shows the formula `=MID(A2,4,6)` in cell Q62. Below the ribbon is a table with 10 rows and 10 columns. The columns are labeled: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Price Range, Day, Country Code, and Month. The data rows contain product information for various items like Laptops, Sneakers, Coffee Makers, Smartphones, Backpacks, Headphones, T-shirts, Blenders, and Tablets. Below the main table is a separate list of months: JAN-US, FEB-US, MAR-US, APR-US, MAY-US, JUN-UK, JUL-UK, AUG-UK, and SEP-UK.

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	\$1,000	30	Electronics	High Price	28	US	JAN-US
3	15-FEB-US	Sneakers	Nike	\$80	15	Fashion	Standard Price	15	US	FEB-US
4	03-MAR-US	Coffee Maker	Keurig	\$130	40	Kitchen	Standard Price	03	US	MAR-US
5	11-APR-US	Smartphone	Samsung	\$900	25	Electronics	High Price	11	US	APR-US
6	22-MAY-US	Backpack	North Face	\$70	20	Outdoor	Standard Price	22	US	MAY-US
7	07-JUN-UK	Headphones	Sony	\$200	45	Electronics	Standard Price	07	UK	JUN-UK
8	19-JUL-UK	T-shirt	Adidas	\$30	5	Fashion	Standard Price	19	UK	JUL-UK
9	23-AUG-UK	Blender	Ninja	\$90	35	Kitchen	Standard Price	23	UK	AUG-UK
10	05-SEP-UK	Tablet	Apple	\$500	50	Electronics	High Price	05	UK	SEP-UK

Month
JAN-US
FEB-US
MAR-US
APR-US
MAY-US
JUN-UK
JUL-UK
AUG-UK
SEP-UK

Answer: Month: JAN-US

Conclusion: In this Price Analysis the above mentioned using all functions in the excel we will get quick analyses of summary, visualizations of table we can easily understand to analysis in product Innovation Hub.